



Maximum sub-array-sum

[Problem](#)[Submissions](#)[Leaderboard](#)

RATE THIS CHALLENGE



Given an array of n integers $x_1, x_2, \dots, x_n$, our task is to find the maximum sub array sum, i.e., For example, in the array $[-1, 2, 4, -3, 5, 2, -5, 2]$ the following subarray produces the maximum sum 10: $[-1, 2, 4, -3, 5, 2, -5, 2]$

Input Format

Two line input First line should read an integer value which represents size of array Second line consists list of elements(integers) separated by space

Constraints

1

Output Format

It prints integer value which represents maximum sub array sum

Sample Input 0

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 0

```
10
```

Sample Input 1

```
7
-3 4 -1 2 1 -4 3
```

Sample Output 1

```
6
```

Sample Input 2

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 2

```
10
```



Sample Input 3

6
-1 2 3 4 5 -15

Sample Output 3

14

Sample Input 4

7
-3 4 -1 2 1 -4 3

Sample Output 4

6

Sample Input 5

5
1 2 3 4 -10

Sample Output 5

10

[Change Theme](#)

Language

C



```
1  #include <stdio.h>
2  #include <string.h>
3  #include <math.h>
4  #include <stdlib.h>
5  int maxsubarr(int arr[],int n){
6      int currmax=arr[0];
7      int totalmax=arr[0];
8      for(int i=1;i<n;i++){
9          if(arr[i]>currmax+arr[i]){
10             currmax=arr[i];
11         }
12         else{
13             currmax+=arr[i];
14         }
15         if(totalmax<currmax){
16             totalmax=currmax;
17         }
18     }
19     return totalmax;
```

```
20 }
21 int main() {
22     int n,arr[50];
23     scanf("%d",&n);
24     for(int i=0;i<n;i++){
25         scanf("%d",&arr[i]);
26     }
27     printf("%d",maxsubarr(arr,n));
28     return 0;
```

Line: 30 Col: 1

 Upload Code as File☐

Test against custom input

Run Code

Submit Code

[Blog](#) | [Scoring](#) | [Environment](#) | [FAQ](#) | [About Us](#) | [Helpdesk](#) | [Careers](#) | [Terms Of Service](#) | [Privacy Policy](#)